

PROJECT		JOB NO.				
CONTRACTOR :						
SUB-CONTRACTOR : Winner Engineering Pte Ltd						
TRADE		DRAWING NO :				
FUNCTIONAL- TESTING COMMISSIONING CHECKLIST (AHU / PAHU)						
Unit Designation :						
Unit Make :						
Model no.						
Location :						
Serial no :						
S/N	Section	Inspection / Test item	CHECKED CONDITION (Please tick inside)			Remarks
			Yes	No	NA	
1	General	Check that the equipment is clean, free of debris, and undamaged before commencing operations.				
2	Documents	Verify chiller tag, make, model, serial number, refrigerant and rated capacity.				
3	B. General installation	Check AHU casing and joints for visible air leakage.				
4	C. Coil, filters & drainage	Check motorized control valve installation, actuator and valve stroke.				
5	C. Coil, filters & drainage	Check filter differential-pressure gauge/switch and tubing.				
6	D. Fan & mechanical	Check fan and motor bearings for abnormal noise, overheating or vibration.				
7	D. Fan & mechanical	Verify fan and motor rotation direction.				
8	D. Fan & mechanical	Check vibration isolators and flexible connections during operation.				
9	E. Dampers & airside	Check fresh-air, return-air, exhaust-air and mixing dampers.				
10	E. Dampers & airside	Check motorized damper actuator, linkage, end switches and command response.				
11	E. Dampers & airside	Check non-return/backdraft damper operation where applicable.				
12	E. Dampers & airside	Check VAV boxes and terminal units are operational and accessible.				
13	F. Electrical & controls	Verify incoming and outgoing power supplies and isolators.				
14	F. Electrical & controls	Perform LCP lamp test and verify Start/Stop/Trip/Run indications.				
15	F. Electrical & controls	Verify Local/Off/Remote or Manual/Auto selector operation.				
16	F. Electrical & controls	Verify VSD operation, frequency command, output frequency and running current.				
17	F. Electrical & controls	Measure motor voltage and running current on all phases.				
18	F. Electrical & controls	Check phase sequence, phase imbalance and motor rotation.				
19	G. Controls, BMS & safety	Verify BMS points against actual field readings and statuses.				
20	G. Controls, BMS & safety	Verify supply-air, return-air, room-air and off-coil temperature sensors.				
21	G. Controls, BMS & safety	Verify duct static-pressure sensor and control setpoint.				
22	G. Controls, BMS & safety	Verify differential-pressure switches for filter and fan status.				
23	G. Controls, BMS & safety	Test fan start/stop, proof-of-flow and fail-to-start/trip alarms.				
24	G. Controls, BMS & safety	Test fire alarm/smoke detector shutdown and damper interlocks where applicable.				

25	G. Controls, BMS & safety	Test freeze-stat/low-temperature protection where applicable.				
26	G. Controls, BMS & safety	Verify occupancy schedule, time clock and after-hours operation.				
27	G. Controls, BMS & safety	Verify temperature setpoint and control sequence, including valve modulation.				
28	H. Functional performance	Record CHW supply and return temperatures during stable operation.				
29	H. Functional performance	Record CHW supply and return pressures or pressure differential.				
30	H. Functional performance	Measure main-duct airflow and compare with design airflow.				
31	H. Functional performance	Measure terminal/grille/diffuser airflow and complete air balancing.				
32	H. Functional performance	Record supply, return, fresh-air and off-coil temperatures and room RH where required.				
33	H. Functional performance	Measure external static pressure and fan total/static pressure as required.				
34	H. Functional performance	Check operating noise and vibration.				
35	H. Functional performance	Run AHU continuously and confirm stable operation.				
36	I. Completion & handover	Confirm all defects/punch-list items are closed or recorded with responsible party.				

Comments to follow up Action :

Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.

CHECKED BY:		WITNESSED BY	INSPECTED BY
NAME			
DESIGNATION :			
COMPANY :	Winner Engineering Pte Ltd		
SIGNATURE :			
DATE :			